Bank Telemarketing Analysis Report

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1. Research Problem and Questions

1.1. Problems

While telemarketing is a primary channel for promoting bank financial products, the current "one-size-fits-all" blind calling strategy is highly inefficient. Our dataset reveals an 88% rejection rate, which means that untargeted campaigns waste labor costs and risk customer fatigue.

Furthermore, analyzing all customers as a single entity masks crucial behavioral differences. A young professional with a mortgage naturally responds to different marketing triggers than a wealthy retiree. Broad data trends fail to tell the sales team exactly when and how (e.g., mobile vs. landline) to contact specific customer types.

In order to solve this business pain point, we put forward a two-step analysis framework:

Step 1: Clustering: First, according to the customer's age, occupation, deposit balance and other characteristics, divide tens of thousands of messy customers into several specific groups with similar characteristics.

Step 2: Association Rules: Then, within each specific small group, we separately explore the specific conditions and combinations that can lead to successful sales (that is the result y = yes).

The ultimate purpose of this project is to provide the sales team with intuitive, data-driven actionable rules (e.g., "If dealing with a high-net-worth retiree, contact them via mobile in October") to maximize conversion rates and eliminate wasteful calls.

Although this two-step framework offers good interpretability and actionability, it mainly explains customers' behavior patterns. To further enhance the predictive performance and better support the decision-making, we further extend our study by introducing supervised learning methods. The neural network method is especially applied to model complex nonlinear functions and improve the subscription prediction accuracy.

1.2. Research Question and Hypothesis

The major research objective is to investigate the elements that influence a client's decision to subscribe to a term deposit (variable y) in a direct marketing campaign. While the classification goal is to forecast subscription results, the overall study problem is to determine which client traits, campaign aspects, and contact techniques have a substantial impact on conversion probability.

Specifically, the following research questions will be investigated:

1. Which types of data, such as demographics (age, employment, marital status), financial indicators (balance, loan status), or campaign-related variables (number of contacts, prior outcomes), have the most predictive power?
2. Does regular interaction lead to higher subscription rates, or are there diminishing returns?
3. How do campaign-related variables affect conversion rates, and can it predict subscription outcomes?

Besides, the primary hypothesis is that campaign-related behavioral variables will have a greater predictive impact than static demographic characteristics. Instead of focusing solely on maximizing forecast accuracy, the research tries to educate on more effective targeting techniques and enhance marketing resource allocation.

2. Data

2.1. Data Introduction

The Bank Marketing dataset is from *UC Irvine Machine Learning Repository*. This whole dataset is based on phone calls and subscription results of the clients which relate directly to the marketing campaigns of a banking institution in Portugal.

The dataset contains 17 features and 45211 samples. Target variable has been recorded as **y**. This dataset contains missing values, which have been recorded as "unknown" in variables “contact” and “poutcome”. Moreover, “job” and “education” variables treat “unknown” as one of its categories. The variable “pdays” contains a coded value “−1” representing clients who were not previously contacted. Although not recorded as a formal missing value, this represents a structurally missing observation because the number of days since last contact is undefined.

Variable Name	Type	Description
age	integer	age
job	categorical	type of job
marital	categorical	marital status
education	categorical	education level
default	binary	whether credit in default

balance	integer	average yearly balance
housing	binary	whether housing loan
loan	binary	whether personal loan
contact	categorical	contact communication type
day_of_week	date	last contact day of the week
month	date	last contact month of year
duration	integer	last contact duration, in seconds
campaign	integer	number of contacts performed during this campaign and for this client
pdays	integer	number of days that passed by after the client was last contacted from a previous campaign
previous	integer	number of contacts performed before this campaign and for this client
poutcome	categorical	outcome of the previous marketing campaign
y	binary	whether the client subscribed a term deposit

Table 1: Variable Introduction

2.2. Data Preparation

To prepare the data, we did the following steps:

1. Parsed the raw semicolon‑separating file into a 17 feature table with 45,211 observations
2. Ensured all string fields are standardized as lowercase strings (i.e., trimmed and lower-cased) to avoid creating duplicated categories due to lack of formatting consistency.
3. Cast numeric variables (age, balance, day, campaign, pdays, previous) to numeric types.
4. Encoded binary variables (default, housing, loan, target y) from “yes/no” to 1/0 to enable modeling and interpretation downstream.
5. This dataset had missing values recorded as “unknown” for several categorical attributes. We treated “unknown” as a valid category for job and education as documented for this dataset, but converted “unknown” to missing (NaN) for the remaining categorical attributes.
   Then we saw high missingness for poutcome and contact. Although poutcome had 36,959 missingness, we carefully considered its use in modeling. Next, we kept contact as an important channel for the campaign and filtered out the contact missing value.
6. We kept the special coding of pdays = −1 as “not previously contacted,” since it represented a customer history state, though this was a conventional missing value.

7. Downloaded the cleaned Bank Marketing dataset for performing clustering analysis, within-cluster association rule discovery, and further prediction of neural networks.

2.3. Why the Data is suitable

The dataset used in the study is well suited to solve the research problem. Real world telemarketing campaigns performed by a banking institution are reflected in the dataset. The number of observations in the dataset is greater than 45,000, which gives a sufficiently large sample size to perform statistical analysis and machine learning. This quantity helps to reduce the overfitting risk and to detect significant customer behavior patterns.

Apart from the large number of observations, the variables in the dataset are also diverse and reflect various aspects of the customer and campaign process. They can be categorized into demographic variables (e.g., age, job, and marital status), financial variables (e.g., balance and presence of loan), and number of contacts and results from previous campaigns. With these features, the study can not only explore who the customers are, but also how they go through the marketing campaign.

On the one hand, the dataset is practical. The data come from real marketing campaigns and the results of the study can be applied to real business scenarios. On the other hand, the analysis does not use variables, e.g., call duration, which are only available after contacting the customer. Only information available before contacting the customer is used, which makes the results more practical. That is, the model can be applied to support business decisions before contacting the client.

3. Choice of Analytical Techniques

To solve the research problem, the following methods are used in the study: unsupervised learning methods (clustering), rule-based learning (association rules) and supervised learning methods (neural network). This study uses these methods in combination in order to capture both the underlying structure in the data and the predictive relations between variables and the target.

Clustering methods are used firstly to group customers according to their similarity. Customer profiles in the data are very diverse, and it is beneficial to group customers together and analyze them as homogeneous groups rather than analyzing the whole population as one single group. Association rule mining is applied in each cluster for y = 1. In this way, meaningful combinations of features are found. These rules are meaningful and easy to interpret. For example, what conditions should be met in order to have a customer answer positively to our marketing campaign.

Finally, the neural network is applied as a predictive method. Since the relationship between variables is modeled as a neural network, it is possible to capture the nonlinear relationships and variable interactions. Compared with linear models, this model is more appropriate for the complexity in the data. Because the target is to find the most suitable customers for each campaign, accurate predictions are of great importance.

4. Why These Methods Fit

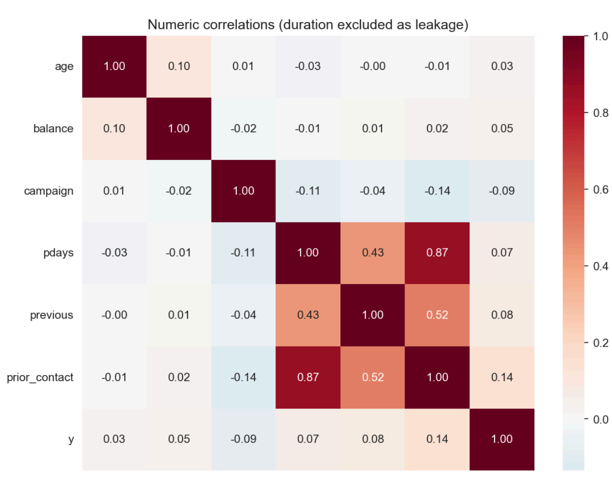
The chosen analytical techniques are appropriate to the aims of this study because the combination of techniques considers both interpretability and predictive aspects of the problem. Customer behavior in a marketing campaign is heterogeneous; that is, different groups of customers may respond differently to marketing. Thus, clustering is a suitable first step in which similar customer groups are identified. This can lead to a better understanding of behavior and a reduction in the dimensionality of the problem.

Association rule mining makes the results more interpretable by identifying specific features combinations that are associated with good behavior within a cluster. Instead of just identifying which groups perform better, this method identifies specific patterns of which groups of customer attributes and which conditions of a marketing campaign are more likely to lead to successful outcomes.

Alternatively, if one of the main goals of analysing the data is to predict whether customers will subscribe to a service then the focus would be on predictive performance. Accurately predicting whether a customer will subscribe is important because one of the aims of this study is to improve the effectiveness of targeting customers. Neural networks are suitable for this purpose because there are nonlinear relationships and complex interactions between the variables that are likely to be difficult to model with simpler techniques.

All in all, this combination of methods focuses on both interpretability and prediction. The combination approach ensures that the results are not only predictive, but also interpretable and meaningful.

5. EDA



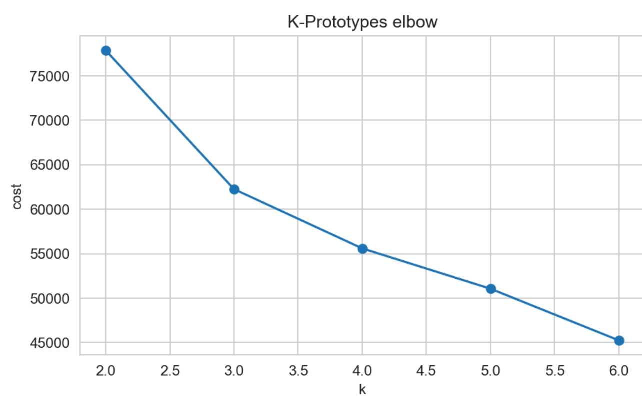
To identify the most predictive variables, we first examined the correlation between numerical features and the target variable y, as shown in the correlation heatmap. It shows that the correlations are relatively weak across all variables, suggesting that no single numerical feature alone strongly determines the outcome. However, campaign-related and behavioral variables show relatively stronger associations with subscription compared to demographic and financial variables.

To address other research problems, we applied two analytical approaches: a rule-based classification model and a neural network model. The rule-based classification model focuses on generating simple and interpretable decision rules, while the neural network model aims to capture more complex patterns in the data and improve predictive performance, especially for the minority class.

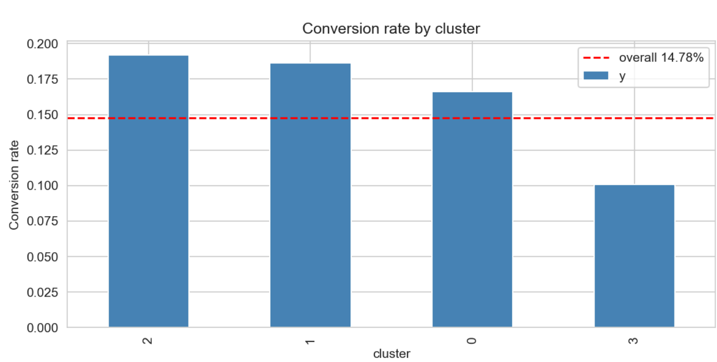
Both methods were evaluated in terms of their predictive performance, particularly their ability to identify potential subscribers, as well as their usefulness for marketing decision-making.

5.1. Rule-based classification model

The rule-based classification model begins with customer segmentation using the K-Prototypes algorithm. To determine the appropriate number of clusters, we applied the elbow method. As shown in the K-Prototypes elbow plot, the cost decreases significantly as k increases from 2 to 4, suggesting that k = 4 is a reasonable choice.



Using k = 4, customers are divided into four clusters, with significantly different conversion rates. Specifically, cluster 2 shows the highest conversion rate of around 19%, followed by clusters 1 and 0, both above the overall average of 14.78%. Cluster 3 has the lowest conversion rate at approximately 10%. These results suggest that the segmentation is meaningful. Certain customer groups are much more likely to subscribe than others. In particular, clusters 1 and 2 represent high-potential segments, while cluster 3 represents a low-response segment.



After clustering, association rules were generated within each cluster, with poutcome added. The per-cluster rules provide more detailed and interpretable patterns behind conversion. For example, in one high-performing cluster, rules involving **poutcome = success and housing = 0** reach a confidence of 76.47% with a lift of 5.16, indicating a very strong association with subscription. In Cluster 1 and Cluster 3, recurring rule patterns include **previously_contacted = 1, housing = 0, loan = 0, and contact = cellular**, with confidence levels around 42% – 44% and lift values close to 3. These results suggest that prior successful engagement and lower household debt burden are important signals of future subscription, but the specific combinations vary across customer groups.

To provide a benchmark, a logistic regression model was built, which achieves a 5-fold cross-validated AUC of 0.747 +/- 0.006, indicating that even a relatively simple linear model can capture meaningful patterns in the data. Examining the model coefficients highlights the importance of key variables. In particular, poutcome shows the strongest positive effect, while factors such as being retired or a student are positively associated with subscription, and housing loans and higher campaign contact frequency are negatively associated, consistent with the rule-based analysis.

To translate the analytical results into business value, we evaluated the top association rule within each cluster and estimated its impact on targeting performance. The results show that the selected rules effectively identify high-response segments within each cluster, achieving higher conversion rates while targeting only a small portion of customers. This indicates more efficient use of marketing resources. Differences across clusters also suggest a trade-off between targeting precision and coverage, allowing campaigns to be adjusted based on business priorities.

cluster	size	base_rate	targeted_size	targeted_conv_rate	lift_vs_cluster	lift_vs_overall	call_volume_pct
0	1288	0.148	51	0.765	5.157	6.537	0.040
1	12119	0.135	1117	0.441	3.261	3.773	0.092
3	13948	0.141	1010	0.433	3.068	3.699	0.072

5.2. Neural network model

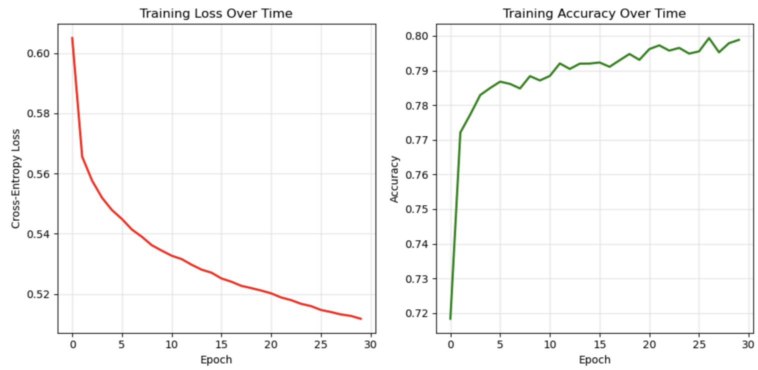
According to the data preparation, the target variable y is highly imbalanced, with approximately 89% of observations labeled as "no" in a dataset of around 40,000 samples. To address this issue, a stratified train-test split was applied to preserve the class distribution in both training and testing sets. In addition, numerical features were standardized, and categorical variables were encoded to ensure that all inputs could be effectively used in the model.

5.2.1. Keras Model with a 2 layers

We first implemented a neural network using Keras with a two-layer architecture. The hidden layer uses ReLU activation to capture non-linear relationships, while the output layer applies a Sigmoid function to produce probabilities for the binary outcome.

From the training process, the loss curves show a steady decrease for both training and validation sets, indicating that the model is learning effectively. The accuracy curves gradually stabilize, and the AUC curve shows consistent improvement, suggesting that the model is able to distinguish between the two classes reasonably well without severe overfitting.

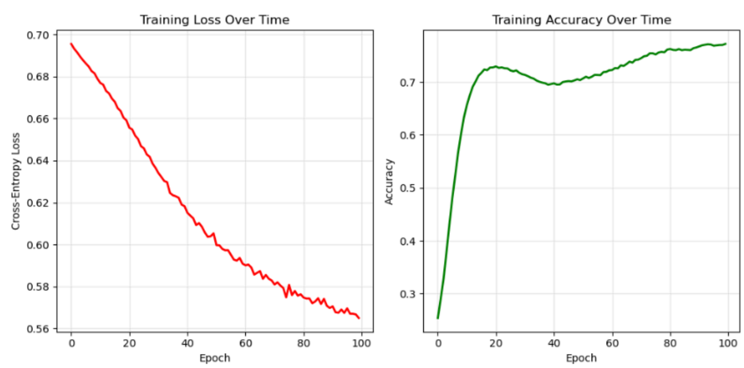
In terms of performance, this model achieves an ROC-AUC of 0.79 and an overall accuracy of 0.80. For the minority "Yes" class, recall reaches 0.62 and the F1-score is 0.46, indicating that the model is able to identify a meaningful portion of actual subscribers. However, precision remains relatively low at 0.36, suggesting that many predicted positive cases are false positives.



5.2.2. PyTorch Model with 3 layers

To further explore performance, we implemented a deeper neural network using PyTorch with 3 layers. This architecture allows the model to capture more complex interactions among variables. The training curves show a steady decrease in loss and a clear increase in accuracy over time, indicating stable learning behavior.

The model achieves an overall test accuracy of 80.48%, indicating acceptable general predictive performance. However, due to class imbalance, class-specific metrics are more informative. The model performs significantly better on the majority "No" class than on the minority "Yes" class. For the "Yes" class, recall reaches 0.64, meaning that a moderate proportion of actual subscribers are identified, but precision is only 0.32, indicating that many predicted positives are incorrect.



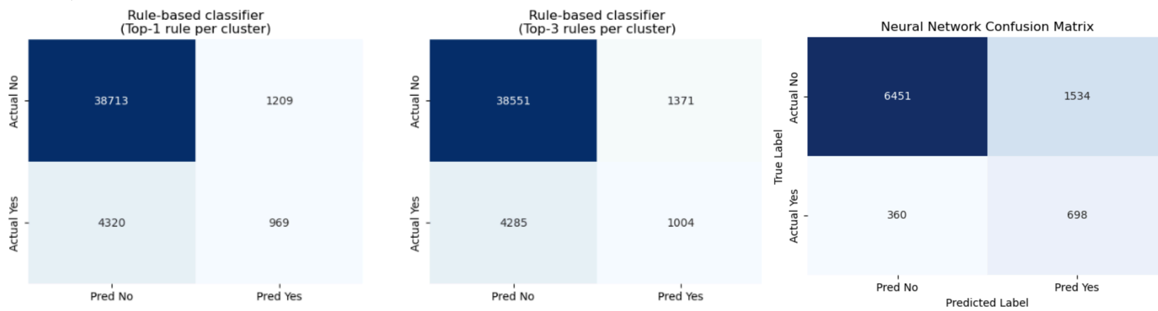
5.2.3. Neural Networking summary

Both neural network models show similar overall performance, with accuracy around 80% and moderate ability to detect the minority class. The Keras model demonstrates slightly more stable training behavior and better overall balance across metrics, supported by its AUC performance and smoother validation curves. The PyTorch model, while slightly deeper, does not provide a clear improvement in predictive performance and still struggles with low precision for the "Yes" class. Among the two, the Keras model is preferred due to its more stable performance and slightly stronger overall results.

5.3. Method Comparison

The rule-based classifier achieves higher overall accuracy (around 88%) by correctly classifying the majority "No" class. However, it performs poorly in detecting the minority "Yes" class, with recall below 0.20. This means that most potential subscribers are not identified, limiting its usefulness for targeting.

In contrast, the neural network achieves lower overall accuracy (around 80%) but significantly improves performance on the "Yes" class. The recall increases to 0.66 and the F1-score to 0.42, indicating a much stronger ability to identify potential subscribers, although precision remains relatively low.



Given that the goal of this project is to identify customers likely to respond "Yes", the neural network is the preferred model. The rule-based approach can still be used as a complementary method due to its interpretability, while future improvements should focus on addressing class imbalance to further enhance performance.

6. Conclusion

Firstly, behavioral and campaign-related variables appear to be more informative than demographic or financial factors in predicting customer subscription. In particular, variables related to prior interactions, such as poutcome, consistently emerge as strong predictive signals across different analytical approaches.

The neural network results further suggest that customer subscription behavior cannot be explained by simple individual variables, but instead depends on more complex interaction patterns across multiple features. This is reflected in the model's ability to improve recall for the "Yes" class, indicating that it can identify potential subscribers more effectively.
However, the relatively low precision indicates that predicting subscription remains challenging, and that customer behavior is inherently uncertain. This suggests that marketing strategies should focus on identifying broader groups of potential subscribers rather than relying on highly precise predictions.

7. Recommendations for Decision Makers

Based on the analysis, marketing efforts should prioritize targeting customers with strong behavioral signals, especially those with positive prior interaction outcomes. These customers show a much higher likelihood of subscription and should be the primary focus of outreach campaigns.
In addition, organizations should avoid excessive repeated contact. The results suggest diminishing returns from frequent outreach, so marketing strategies should emphasize quality of targeting rather than quantity of contact, reducing unnecessary calls and improving overall efficiency.

Finally, predictive tools should be used to support customer selection, while interpretable rules can guide business understanding. In practice, this means combining data-driven targeting with clear decision rules to improve both effectiveness and transparency in marketing campaigns.

8. Limitations, Improvements, and Implications

A key limitation of this study is the strong class imbalance, with a large majority of “No” cases. This makes it difficult for models to accurately identify the minority “Yes” class and affects evaluation results, especially precision and recall. In addition, some variables, such as duration, are only available after the interaction, which limits their usefulness for pre-campaign prediction. The neural network model also lacks interpretability, making it harder to explain results to business stakeholders.

Future improvements should focus on better handling class imbalance, such as applying oversampling or other resampling techniques to improve detection of the “Yes” class. Further model tuning or exploring alternative approaches may also enhance performance. In addition, clearly separating variables that are available before and after contact would make the analysis more practical for real-world applications.

The findings imply that marketing strategies should focus on identifying high-potential customers rather than increasing outreach volume. Data-driven approaches can improve targeting efficiency, while combining predictive models with interpretable insights can support more effective and transparent decision-making.

9. References

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